

5 - dars: Logging va Error (Theory + amaliy)
 , Katta file va dataset/ni githubga push qilish
 , Data loading + log (project)

Logging va Error

Kodlar yozilganda turli xil xolat/ bo'ladi:

- ↳ Warnings
- ↳ Errors

Kodlarni nazorat qilishni 2ta asosiy
 yo'li mavjud:

- ↳ Logging
- ↳ Errors (Exceptions)

• Logging => Program run bo'lgunicha
 bo'lgan jarayon/ni (xodisalarni) saqlab bo-
 radigan fayl.

• Foydasi

- ↳ Debug
- ↳ Monitor
- ↳ Audit

Tabirlar turli xil darajada bo'ladi:

DEBUG --- Iltimos haqida chuqurroq ma'lumot
 INFO --- Programma haqida umumiy ma'lumot

No.

→ Nimadir deprecated; eskiygan yoli xavdan oqchlatirish

WARNING -- Kodlar run bōlishda davom etadi, lekin kutilmagan nimadir sodir bōlganini kōrsatadi.

ERROR -- Jiddiy muammo (ma'lum bir block ishlamay qoladi) (yoki model)

CRITICAL -- Programma tōxtaydi juda jiddiy muammo. Eng yuqori darajadagi muammo hisoblanadi.

→ (Chat GPT dan ML uchun kerakli error tōyxatini qidirish)
Errorlar -- runtime problem bu normal bōlgan ishlarni tōxtatadi

- Syntax Error - Code is invalid
- Type Error - wrong type of variable
- FileNotFoundError - missing file
- ValueError - invalid value for a function

Logging va Error (amaliy)

Logging

```
import logging
logging.basicConfig(level=logging.INFO)
logging.info("Model qurishni boshladi")
logging.warning("Bu usul yozilishi mumkin, boshqa variantlarni ko'ring")
```

logging.error ("Xatolik ro'y berdi! Daytadan tekshiring")

Updated version of above code(s)

logging.basicConfig(level=logging.ERROR, force=True)

bu shu yerdan errorni olib ketadi. force=True shunga hech berib qolganlarini chiqarmay faqat shuni qoldiradi.

Error

try:
pass
except:
pass



try :

x = int(3)

if x > 0:

print("x soni musbat")

: else:

print("x soni manfiy yoki nolga teng")

~~except~~ except ValueError as e:

print(f"xatolik sodir boldi: {e}")

logging configuration

logging.basicConfig (

filename = r'copy path of log',
 filemode = 'a', # a - bu append (qo'shib
 level = logging.INFO, # qo'shib
 format = "%(asctime)s - %(levelname)s -
 %(message)s"

logging.info ("Project boshlandi")

↳ ... INFO:root: Project boshlandi

Katta file va datasetlarni githubga
 push qilish

• Githubda limit :

↳ 100 MB xer bir file uchun

↳ Tavsiga qilinadi 1-GB butun repository
 4-h.

• Mceammlar :

↳ Katta filelarni Github reject qiladi.

↳ Katta filelar repositoryni performance
 sekinlashtiradi

↳ Bazi filelarni yozilishiga
 olib keladi.

yechi mlari:

- Git LFS (Git Large File Storage) - track dataset, files
- Cloud Storage for Dataset (pullik \$)
- .gitignore (keep code, data external) (bu datasetni ignore qiladi, qolganlarini push qiladi)

Birinchi navbatda virtual environmentni activate qilishiimiz h-h. `brew install git-lfs`,
`git-lfs install`

```
git lfs track "*.csv"
```

> Tracking "*.csv" } => chiqadi
 (my env)

Hohlasak `.py` filalar ham b'lishi mumkin uni tracking qilishiimiz ham mumkin.

- `ipynb`
- `pkl` (Modelimiz)
- `h5` (DL da o'tamiz ~~ker~~ rasm/ b'u)

`lfs`ni install qilish kerak 4-navbatda

```
git lfs ls-files
```

1. LFS to'g'ri ishlayaptimi, tekshi - ~~ishmagan~~ tekshirib b'lsangiz.
2. yangi `.pkl`, `.csv`, `.h5`, `.pt` kabi fayl qo'shdingiz va u LFS ga tashqarini tekshirib chiqsiz. Shunda ishlatilgan

Git LFS => model, dataset, video, rasm kabi katta binary fayllar uchun mo'ljallangan

- .py → oddiy Git
- README.md → oddiy Git
- requirements.txt → oddiy Git
- .csv (agar juda katta bo'lsa) → Git LFS
- .pkl, .joblib, .pt, .h5 → Git LFS

```
git add.
```



```
git commit -m "Large files updated"
```



```
git push -u origin main
```

Data loading + Log (Project)

Data loading + Log (Bundan oldin logging basic Configni berib o'tirish kerak)

```
data_path = r"Biron bir datasetning copy pathi"
```

try:

```
df = pd.read_csv(data_path)
```

```
logging.info(f"Dataset muammosiz yuklandi,  
va uning o'lchami: {df.shape}")
```

except File Not Found Error as e:

```
logging.error(f"Berilgan manzil bo'yicha  
dataset topilmadi: {data_path}")
```

Biz qilayotgan ishni qayerlariga
qo'shib ketsa bo'ladi.

try :

```
df = pd.concat([df-1, df-2], ignore-index=
logging.info(f"Dataset birlashtirildi") True)
except FileNotFoundError as e:
    logging.error(f"xatolik bor! Ma'lum bir
dataset mavjud emas!")
```

! Important : Bitta jabha bor ekanki
Heu agar X_{train} scaled va X_{test} -
scaledni ~~shunda~~ ^{preprocessed} paplarga saqlasheli
bo'lsan quyidagi kodlarni bayqirishim kerak.

```
import os
```

```
save_dr = "copy path"
```

```
os.makedirs(save_dr, exist_ok = True)
```

```
X_train_scaled.to_csv (
    os.path.join(save_dr, "X_train_scaled.csv"),
    index = True)
```

```
X_test_scaled.to_csv (
    os.path.join(save_dr, "X_test_scaled.csv"),
    index = True)
```